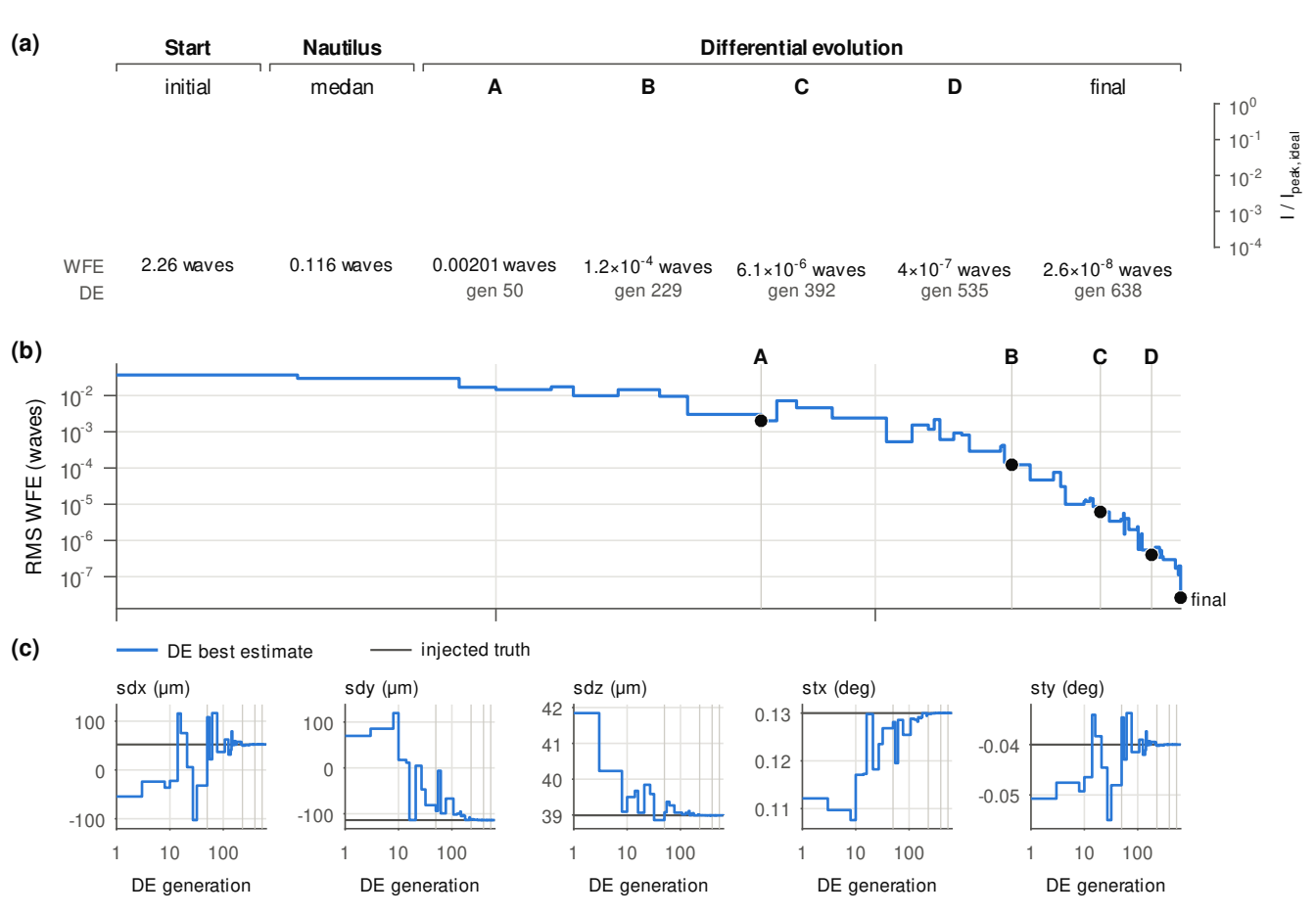
(a)
Start
Nautilus
Differential evolution
initial
medan
A
B
C
D
final
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
$I / I_{peak, ideal}$
WFE
DE
2.26 waves
0.116 waves
0.00201 waves
gen 50
1.2×10^{-4} waves
gen 229
6.1×10^{-6} waves
gen 392
4×10^{-7} waves
gen 535
2.6×10^{-8} waves
gen 638
(b)
RMS WFE (waves)
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
A
B
C
D
final
(c)
DE best estimate
injected truth
sdx (μm)
sdy (μm)
sdz (μm)
stx (deg)
sty (deg)
100
0
-100
42
41
40
39
0.13
0.12
0.11
-0.04
-0.05
1
10
100
DE generation